

# **Лабораторная работа №2**

**Структуры данных**

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# 1 Цель работы

Основная цель работы – изучить несколько структур данных, реализованных в Julia, научиться применять их и операции над ними для решения задач.

## 2 Задание

1. Используя Jupyter Lab, повторите примеры из раздела 2.2.
2. Выполните задания для самостоятельной работы (раздел 2.4).

## 3 Выполнение лабораторной работы

### 3.1 Повтор примеров

Выполним примеры из раздела кортежи [1].

Кортеж (Tuple) – структура данных (контейнер) в виде неизменяемой индексируемой последовательности элементов какого-либо типа (элементы индексируются с единицы).

```
In [19]: ()  
Out[19]: ()  
  
In [20]: favoritelang = ("Python", "Julia", "R")  
Out[20]: ("Python", "Julia", "R")  
  
In [21]: x1 = (1, 2, 3)  
Out[21]: (1, 2, 3)  
  
In [22]: x2 = (1, 2.0, "tmp")  
Out[22]: (1, 2.0, "tmp")  
  
In [23]: x3 = (a=2, b=1+2)  
Out[23]: (a = 2, b = 3)
```

Рисунок 3.1: Кортежи 1



```

In [24]: length(x2)
Out[24]: 3

In [25]: x2[1], x2[2], x2[3]
Out[25]: (1, 2.0, "tmp")

In [26]: c = x1[2] + x1[3]
Out[26]: 5

In [27]: x3.a, x3.b, x3[2]
Out[27]: (2, 3, 3)

In [28]: in("tmp", x2), 0 in x2
Out[28]: (true, false)

```

Рисунок 3.2: Кортежи 2

Выполним примеры из раздела словари.

Словарь – неупорядоченный набор связанных между собой по ключу данных.

```

In [29]: phonebook = Dict{
    "Иванов И.И." => {"867-5309", "333-5544"},
    "Бухгалтерия" => "555-2368"
}
Out[29]: Dict{String, Any} with 2 entries:
  "Бухгалтерия" => "555-2368"
  "Иванов И.И." => {"867-5309", "333-5544"}

In [30]: keys(phonebook)
Out[30]: KeySet for a Dict{String, Any} with 2 entries. Keys:
  "Бухгалтерия"
  "Иванов И.И."

In [31]: values(phonebook)
Out[31]: ValueIterator for a Dict{String, Any} with 2 entries. Values:
  "555-2368"
  {"867-5309", "333-5544"}

In [32]: pairs(phonebook)
Out[32]: Dict{String, Any} with 2 entries:
  "Бухгалтерия" => "555-2368"
  "Иванов И.И." => {"867-5309", "333-5544"}

```

Рисунок 3.3: Словари 1

```

In [33]: haskey(phonebook, "Иванов И.И.")
Out[33]: true

In [34]: phonebook["Сидоров П.С."] = "555-3344"
Out[34]: "555-3344"

In [35]: pop!(phonebook, "Иванов И.И.")
Out[35]: {"867-5309", "333-5544"}

In [36]: a = Dict{"foo" => 0.0, "bar" => 42.0};
          b = Dict{"baz" => 17, "bar" => 13.0};
          merge(a, b), merge(b, a)
Out[36]: {Dict{String, Real}{"bar" => 13.0, "baz" => 17, "foo" => 0.0}, Dict{String, Real}{"bar" => 42.0, "baz" => 17, "foo" => 0.0}}

```

Рисунок 3.4: Словари 2

Выполним примеры из раздела множества.

Множество, как структура данных в Julia, соответствует множеству, как математическому объекту, то есть является неупорядоченной совокупностью элементов какого-либо типа. Возможные операции над множествами: объединение, пересечение, разность; принадлежность элемента множеству.

```

In [37]: A = Set{[1, 3, 4, 5]}
Out[37]: Set{Int64} with 4 elements:
          5
          4
          3
          1

In [38]: B = Set{"abracadabra"}
Out[38]: Set{Char} with 5 elements:
          'a'
          'd'
          'r'
          'k'
          'b'

In [39]: S1 = Set{[1, 2]}
          S2 = Set{[3, 4]}
          issetequal(S1, S2)
Out[39]: false

In [40]: S3 = Set{[1, 2, 2, 3, 1, 2, 3, 2, 1]}
          S4 = Set{[2, 3, 1]}
          issetequal(S3, S4)
Out[40]: true

```

Рисунок 3.5: Множества 1

```

Out[40]: true

In [43]: C = union(S1, S2)
Out[43]: Set{Int64} with 4 elements:
         4
         2
         3
         1

In [44]: D = intersect(S1, S3)
Out[44]: Set{Int64} with 2 elements:
         2
         1

In [45]: E = setdiff(S3, S1)
Out[45]: Set{Int64} with 1 element:
         3

In [46]: issubset(S1, S4)
Out[46]: true

In [47]: push!(S4, 99)
Out[47]: Set{Int64} with 4 elements:
         2
        99
         3
         1

In [48]: pop!(S4)
Out[48]: 2

```

Рисунок 3.6: Множества 2

Выполним примеры из раздела массивы.

Массив – коллекция упорядоченных элементов, размещённая в многомерной сетке. Векторы и матрицы являются частными случаями массивов.

```

In [49]: empty_array = []
Out[49]: Any[]

In [52]: empty_array_2 = (Int64)[]
Out[52]: Int64[]

In [53]: empty_array_3 = (Float64)[]
Out[53]: Float64[]

In [54]: a = [1, 2, 3]
Out[54]: 3-element Vector{Int64}:
 1
 2
 3

In [55]: b = [1 2 3]
Out[55]: 1×3 Matrix{Int64}:
 1  2  3

In [56]: A = [[1, 2, 3] [4, 5, 6] [7, 8, 9]]
Out[56]: 3×3 Matrix{Int64}:
 1  4  7
 2  5  8
 3  6  9

```

Рисунок 3.7: Массивы 1

```

In [57]: B = [[1 2 3]; [4 5 6]; [7 8 9]]
Out[57]: 3×3 Matrix{Int64}:
 1  2  3
 4  5  6
 7  8  9

In [58]: c = rand(1, 8)
Out[58]: 1×8 Matrix{Float64}:
 0.16822  0.278178  0.0879463  0.485416  _  0.777814  0.829817  0.89761
 1

In [59]: C = rand(2, 3)
Out[59]: 2×3 Matrix{Float64}:
 0.187428  0.538214  0.875849
 0.851585  0.472582  0.795135

```

Рисунок 3.8: Массивы 2

```

Out[59]: 2x3 Matrix(Float64):
          0.187428  0.538214  0.875849
          0.851585  0.472582  0.795135

In [62]: D = rand(4, 3, 2)

Out[62]: 4x3x2 Array{Float64, 3}:
[:, :, 1] =
          0.272256  0.638178  0.10435
          0.495959  0.83931  0.845109
          0.0489139  0.141009  0.416999
          0.315541  0.636198  0.140045

[:, :, 2] =
          0.576745  0.00782045  0.179376
          0.854092  0.799871  0.257626
          0.46979  0.694563  0.537992
          0.465904  0.674004  0.334386

In [63]: roots = [sqrt(i) for i in 1:10]

Out[63]: 10-element Vector{Float64}:
          1.0
          1.4142135623730951
          1.7320508075688772
          2.0
          2.23606797749979
          2.449489742783178
          2.6457513110645907
          2.8284271247461903
          3.0
          3.1622776601683795

```

Рисунок 3.9: Массивы 3

```

In [63]: roots = [sqrt(i) for i in 1:10]

Out[63]: 10-element Vector{Float64}:
          1.0
          1.4142135623730951
          1.7320508075688772
          2.0
          2.23606797749979
          2.449489742783178
          2.6457513110645907
          2.8284271247461903
          3.0
          3.1622776601683795

In [64]: ar_1 = [3*i^2 for i in 1:2:9]

Out[64]: 5-element Vector{Int64}:
          3
          27
          75
          147
          243

In [65]: ar_2 = [i^2 for i=1:10 if (i^2%5!=0 && i^2%4!=0)]

Out[65]: 4-element Vector{Int64}:
          1
          9
          49
          81

```

Рисунок 3.10: Массивы 4

```
In [66]: ones(5)
Out[66]: 5-element Vector{Float64}:
 1.0
 1.0
 1.0
 1.0
 1.0
```

```
In [67]: ones(2, 3)
Out[67]: 2x3 Matrix{Float64}:
 1.0  1.0  1.0
 1.0  1.0  1.0
```

```
In [68]: zeros(4)
Out[68]: 4-element Vector{Float64}:
 0.0
 0.0
 0.0
 0.0
```

```
In [69]: fill(3.5, (3, 2))
Out[69]: 3x2 Matrix{Float64}:
 3.5  3.5
 3.5  3.5
 3.5  3.5
```

Рисунок 3.11: Массивы 5

```
In [70]: repeat([1, 2], 3, 3)
Out[70]: 6x3 Matrix{Int64}:
 1  1  1
 2  2  2
 1  1  1
 2  2  2
 1  1  1
 2  2  2
```

```
In [71]: repeat([1 2], 3, 3)
Out[71]: 3x6 Matrix{Int64}:
 1  2  1  2  1  2
 1  2  1  2  1  2
 1  2  1  2  1  2
```

```
In [72]: a = collect(1:12)
Out[72]: 12-element Vector{Int64}:
 1
 2
 3
 4
 5
 6
 7
 8
 9
10
11
12
```

Рисунок 3.12: Массивы 6

```

In [73]: b = reshape(a, (2, 6))
Out[73]: 2×6 Matrix{Int64}:
 1  3  5  7  9 11
 2  4  6  8 10 12

In [74]: b'
Out[74]: 6×2 adjoint{::Matrix{Int64}} with eltype Int64:
 1  2
 3  4
 5  6
 7  8
 9 10
11 12

In [75]: c = transpose(b)
Out[75]: 6×2 transpose{::Matrix{Int64}} with eltype Int64:
 1  2
 3  4
 5  6
 7  8
 9 10
11 12

```

Рисунок 3.13: Массивы 7

```

In [76]: ar = rand(10:20, 10, 5)
Out[76]: 10×5 Matrix{Int64}:
17 15 17 12 10
16 10 20 14 12
16 13 19 16 10
16 18 10 10 10
16 16 15 17 15
17 10 11 10 18
19 13 10 14 11
16 13 20 16 10
14 13 10 18 18
11 16 15 19 17

In [77]: ar[:, 2]
Out[77]: 10-element Vector{Int64}:
15
10
13
18
16
10
13
13
13
16

```

Рисунок 3.14: Массивы 8

```
In [78]: ar[:, [2, 5]]
```

```
Out[78]: 10x2 Matrix{Int64}:  
 15 10  
 10 12  
 13 10  
 18 10  
 16 15  
 10 18  
 13 11  
 13 10  
 13 18  
 16 17
```

```
In [80]: ar[:, 2:4]
```

```
Out[80]: 10x3 Matrix{Int64}:  
 15 17 12  
 10 20 14  
 13 19 16  
 18 10 10  
 16 15 17  
 10 11 10  
 13 10 14  
 13 20 16  
 13 10 18  
 16 15 19
```

```
In [81]: ar[[2, 4, 6], [1, 5]]
```

```
Out[81]: 3x2 Matrix{Int64}:  
 16 12  
 16 10  
 17 18
```

Рисунок 3.15: Массивы 9



```
In [82]: ar[1, 3:end]
Out[82]: 3-element Vector{Int64}:
 17
 12
 10
```

```
In [83]: sort(ar, dims=1)
Out[83]: 10x5 Matrix{Int64}:
 11 10 10 10 10
 14 10 10 10 10
 16 13 10 12 10
 16 13 11 14 10
 16 13 15 14 11
 16 13 15 16 12
 16 15 17 16 15
 17 16 19 17 17
 17 16 20 18 18
 19 18 20 19 18
```

```
In [84]: sort(ar, dims=2)
Out[84]: 10x5 Matrix{Int64}:
 10 12 15 17 17
 10 12 14 16 20
 10 13 16 16 19
 10 10 10 16 18
 15 15 16 16 17
 10 10 11 17 18
 10 11 13 14 19
 10 13 16 16 20
 10 13 14 18 18
 11 15 16 17 19
```

Рисунок 3.16: Массивы 10

```
In [85]: ar .> 14
Out[85]: 10x5 BitMatrix:
 1 1 1 0 0
 1 0 1 0 0
 1 0 1 1 0
 1 1 0 0 0
 1 1 1 1 1
 1 0 0 0 1
 1 0 0 0 0
 1 0 1 1 0
 0 0 0 1 1
 0 1 1 1 1
```

Рисунок 3.17: Массивы 11

```

In [86]: findall(ar .> 14)
Out[86]: 27-element Vector{CartesianIndex{2}}:
 CartesianIndex{1, 1}
 CartesianIndex{2, 1}
 CartesianIndex{3, 1}
 CartesianIndex{4, 1}
 CartesianIndex{5, 1}
 CartesianIndex{6, 1}
 CartesianIndex{7, 1}
 CartesianIndex{8, 1}
 CartesianIndex{1, 2}
 CartesianIndex{4, 2}
 CartesianIndex{5, 2}
 CartesianIndex{10, 2}
 CartesianIndex{1, 3}
 ⋮
 CartesianIndex{5, 3}
 CartesianIndex{8, 3}
 CartesianIndex{10, 3}
 CartesianIndex{3, 4}
 CartesianIndex{5, 4}
 CartesianIndex{8, 4}
 CartesianIndex{9, 4}
 CartesianIndex{10, 4}
 CartesianIndex{5, 5}
 CartesianIndex{6, 5}
 CartesianIndex{9, 5}
 CartesianIndex{10, 5}

```

Рисунок 3.18: Массивы 12

## 3.2 Задания для самостоятельной работы

*Задание 1.* Работа со множествами.

```

In [88]: A = Set{[0, 3, 4, 9]}
         B = Set{[1, 3, 4, 7]}
         C = Set{[0, 1, 2, 4, 7, 8, 9]}
         P = intersect(union(A, B), union(A, B), union(A, C), union(B, C))
Out[88]: Set{Int64} with 6 elements:
 0
 4
 7
 9
 3
 1

```

Рисунок 3.19: Работа со множествами

*Задание 2.* Операции над множествами элементов разных типов.

```

In [99]: aaa = (12, 13, 14); bbb = [444, 4421, 3456]
length(aaa)

Out[99]: 3

In [100]: ccc = [aaa[end], pop!(bbb), "RRR"]

Out[100]: 3-element Vector{Any}:
           14
          3456
          "RRR"

In [102]: ddd = [i for i in ccc]

Out[102]: 3-element Vector{Any}:
           14
          3456
          "RRR"

```

Рисунок 3.20: Операции над множествами элементов разных типов

Задание 3. Создание разными способами массивов и векторов.

Задание 3.1.

```

ar = [i for i in 1:21]
print(ar)

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20,
 21]

```

Рисунок 3.21: Задание 3.1

Задание 3.2.

```

ar = reverse([i for i in 1:21])
print(ar)

[21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2,
 1]

ar = [i for i in 21:-1:1]
print(ar)

[21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2,
 1]

```

Рисунок 3.22: Задание 3.2

Задание 3.3.

```
ar = vcat(1:21, 20:-1:1)
print(ar)

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20,
21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2,
1]
```

Рисунок 3.23: Задание 3.3

Задание 3.4.

```
tmp = [4 6 3]
print(tmp)

[4 6 3]
```

Рисунок 3.24: Задание 3.4

Задание 3.5.

```
ar = vcat([tmp[1] for _ in 1:10], tmp[2:end])
print(ar)

[4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 6, 3]
```

Рисунок 3.25: Задание 3.5

Задание 3.6.

```
ar = vcat([tmp[1] for _ in 1:10], [tmp[2] for _ in 1:10],
[tmp[3] for _ in 1:10])
print(ar)

[4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 3, 3, 3, 3,
3, 3, 3, 3, 3, 3]
```

Рисунок 3.26: Задание 3.6

Задание 3.7.

```
ar = vcat([tmp[1] for _ in 1:11], [tmp[2] for _ in 1:10],  
          [tmp[3] for _ in 1:10])  
print(ar)
```

[4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 3, 3, 3,  
 3, 3, 3, 3, 3, 3, 3]

Рисунок 3.27: Задание 3.7

Задание 3.8.

```
ar = vcat([tmp[1] for _ in 1:10], [tmp[2] for _ in 1:20],  
          [tmp[3] for _ in 1:30])  
print(ar)
```

[4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6,  
 6, 6, 6, 6, 6, 6, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,  
 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3]

Рисунок 3.28: Задание 3.8

Задание 3.9.

```
tmp = [4 6 3]  
ar = []  
for i in tmp  
    push!(ar, 2^i)  
end  
ar = vcat(ar[1:2], [ar[3] for _ in 1:4])  
ar = string.(ar)  
ar = join(ar, "")  
counter = count(isequal('6'), ar)  
print(counter)
```

2

Рисунок 3.29: Задание 3.9

Задание 3.10.

53.11374594642971

Рисунок 3.30: Задание 3.10

### Задание 3.11.

```

vect = []
for j in 1:3:34
    i = j + 2
    push!(vect, (0.1i, 0.2j))
end
print(vect)

Any[(0.00100000000000000002, 0.2), (1.0000000000000004e-6, 0.0016000000000000003), (1.0000000000000005e-9, 1.2800000000000005e-5), (1.000000000000006e-12, 1.0240000000000006e-7), (1.0000000000000009e-15, 8.192000000000005e-10), (1.000000000000001e-18, 6.553600000000005e-12), (1.0000000000000012e-21, 5.2428800000000056e-14), (1.0000000000000014e-24, 4.194304000000005e-16), (1.0000000000000015e-27, 3.3554432000000048e-18), (1.0000000000000017e-30, 2.684354560000004e-20), (1.0000000000000018e-33, 2.1474836480000035e-22), (1.000000000000002e-36, 1.717986918400003e-24)]

```

Рисунок 3.31: Задание 3.11

Задание 3.12.

```

vect = []
for i in 1:25
    push!(vect, (2^i)/i)
end
print(vect)

Any[2.0, 2.0, 2.6666666666666665, 4.0, 6.4, 10.666666666666666, 18.2857
14285714285, 32.0, 56.888888888888886, 102.4, 186.1818181818182, 341.33
3333333333, 630.1538461538462, 1170.2857142857142, 2184.5333333333333,
4096.0, 7710.117647058823, 14563.555555555555, 27594.105263157893, 5242
8.8, 99864.38095238095, 190650.18181818182, 364722.0869565217, 699050.6
666666666, 1.34217728e6]
```

Рисунок 3.32: Задание 3.12

Задание 3.13.

```

vect = []
for i in 1:30
    push!(vect, "fn$i")
end
print(vect)

Any["fn1", "fn2", "fn3", "fn4", "fn5", "fn6", "fn7", "fn8", "fn9", "fn10", "fn11", "fn12", "fn13", "fn14", "fn15", "fn16", "fn17", "fn18", "fn19", "fn20", "fn21", "fn22", "fn23", "fn24", "fn25", "fn26", "fn27", "fn28", "fn29", "fn30"]

```

Рисунок 3.33: Задание 3.13

#### Задание 3.14.

```

vect = []
for i in 1:249
    j = i + 1
    push!(vect, y[j] - x[i])
end
print(vect)

Any[281, 59, 188, -470, 587, -699, 370, 126, -434, -96, -374, 771, 449, -126, 892, 98, -568, 221, -129, -687, -313, 786, -8, 192, -333, -190, -389, 350, -502, 203, -265, -948, -27, 200, 82, -351, -299, -175, 232, -481, -506, -642, -47, -459, -583, 132, 379, -228, 294, -217, 512, -269, -372, 169, -287, -216, 357, -279, -257, -117, -21, -109, 480, -498, 697, 295, 23, 693, 115, -44, 362, -259, 179, -407, 66, 175, 7, 250, 398, -708, 82, -145, -190, 548, -37, -202, -43, -164, 87, 571, 390, -424, -267, -308, -599, -749, 91, -809, 50, 134, -55, 66, 668, 317, 547, 572, 231, 162, -445, 180, 191, 182, -394, 110, -158, 825, -46, 569, -576, 515, -128, -234, -659, -69, -455, 277, -189, 738, 34, 515, 178, -508, -374, 248, -237, 430, 373, -100, 38, 58, 705, -248, -143, 127, -74, -575, 673, -401, -554, 47, -334, -309, -615, 146, 23, -517, -476, -767, 63, -853, -140, 377, 144, -616, -399, 591, 406, -544, 521, 654, 823, -250, 427, 405, 254, 158, -169, 703, -94, 370, -402, 450, 118, -202, 25, -45, -871, -913, -208, -563, -646, 798, 116, -38, 205, 230, 464, -252, -274, -929, 657, -350, 98, -360, 66, -306, 123, -88, -59, 83, 0, 424, 480, -93, 22, 35, -565, -337, -279, 347, 592, -489, -96, -619, -391, -212, -394, -488, 655, 517, -106, -3, 421, -741, -3, -258, 415, 67, 38, 365, 748, 288, -873, -351, 329, 906, 537, -535, -125]

```

Рисунок 3.34: Задание 3.14.1

```

vect = [x[i] + 2*x[i+1] - x[i+2] for i in 1:248]
print(vect)

```

[1824, 79, 1735, 641, 1855, 1690, 44, 1030, 1297, 2210, 687, 663, 1235, 196, -23, 968, 1404, 179, 1776, 1952, 253, 342, 1001, 270, 1236, 2609, 198, 1121, 1697, 687, 2190, 1104, 394, 429, 630, 1093, 1046, 388, 1043, 1676, 2621, 655, 677, 2094, 1610, 235, 1862, 1534, 910, 17, 708, 1085, 523, 1753, 1556, -432, 1287, 1055, 1150, 1005, 464, -364, 1555, 791, -585, 1183, 442, 979, 1571, 578, 899, 1471, 1255, 2029, 898, 1684, 1150, -114, 1192, 722, 1232, 2577, 760, 874, 1443, 761, 1011, 75, 454, 506, 1181, 1124, 1598, 1920, 2385, 496, 1758, 700, 660, 1006, 497, -127, 344, 744, 326, 1265, 750, 1619, 531, 1614, 376, 332, 1357, 2134, 131, 1568, 415, 1760, 1238, 1081, 1564, 2257, 649, 755, 1473, 1773, -169, 1395, 605, 428, 816, 1845, 98, 1667, 378, 338, 1126, 1426, 926, 194, 801, 1526, 1547, 1299, 1691, 569, 201, 1700, 1367, 1245, 1659, 1692, 1071, -51, 1455, 1378, 1697, 1380, 2168, 1541, 780, 825, 1994, 1614, 1194, -219, 830, 1455, 358, -48, 583, 750, 796, 126, -275, 1664, 690, 761, 135, 1372, -258, 991, 1545, 1668, 1345, 1649, 2295, 1112, 1066, 2354, 857, 259, 337, 683, 543, 852, 250, 747, 2412, 528, 1340, 1117, 624, 805, 1495, 1455, 184, 1921, 395, 934, 1142, -66, 839, 1559, 50, 877, 1184, 1858, 911, -706, 1412, 461, 1295, 2093, 1201, 440, 2382, 771, 213, 352, 315, 397, 1857, 979, 1660, 726, 606, 1893, 1171, 353, -425, 1569, 2003, 1077, 362, -896, 1760, 647]

Рисунок 3.35: Задание 3.14.2

```

vect = [round(sin(y[i])/cos(x[i+1])), digits=2] for i in 1:249]
print(vect)

```

[-0.07, 0.92, 1.06, -1.49, 9.04, 1.91, -0.16, -0.38, 0.91, 0.86, 1.69, 0.06, -0.49, -0.88, 0.42, 0.36, 4.1, 0.5, -0.77, 0.94, -1.53, 1.36, 1.0, -0.83, -0.69, -0.52, -1.12, -1.07, 1.29, -0.14, -0.8, 0.87, -3.14, -1.46, -6.19, -0.57, -0.94, -0.28, 0.36, 1.22, -1.82, 0.62, -1.54, -0.86, 2.7, 3.49, 3.0, 4.84, 0.58, -0.31, 0.18, 0.76, 1.06, 0.99, 1.16, -1.16, 1.35, 3.29, -2.79, 0.83, -1.28, 1.34, 1.32, -0.9, 5.09, 0.64, -1.61, 0.87, 1.17, -6.42, -1.65, 0.69, -0.48, 0.32, -0.68, -1.39, 3.48, -4.58, 0.72, -3.29, -1.04, 0.98, -9.96, 1.08, -0.55, 0.9, 1.01, -2.42, 0.54, -0.95, 0.9, 0.86, 0.31, 0.31, 0.11, 0.57, -1.12, 0.17, -0.94, -5.82, -1.25, -0.12, 0.61, -222.03, 0.22, -1.1, 0.83, 1.23, 1.08, 205.09, 2.52, -3.91, 0.96, -0.71, -0.5, 0.07, -3.15, 0.5, -0.83, 1.04, -0.03, -1.09, 1.65, -1.12, -0.89, -1.22, -2.01, -1.73, 0.77, -0.79, 0.33, 1.05, -1.18, 1.05, -0.77, -0.35, -0.62, 11.58, -1.19, 1.63, -0.82, 2.52, -2.03, 0.57, -0.05, 0.98, 0.67, -0.22, -1.12, 213.76, -3.3, 4.99, -0.59, 1.04, 1.63, -0.11, 0.45, 1.01, 0.38, -0.06, 0.74, -2.91, -1.33, -4.69, 1.14, -0.99, 1.03, -5.13, -0.63, -1.03, -0.84, -0.89, 1.55, 0.67, 1.06, -0.44, -1.25, 0.94, -1.39, -1.39, -1.44, -1.44, 3.75, 0.88, 0.39, 1.377, -0.7, -1.78, 0.96, -0.52, -0.7, -1.0, 0.69, -0.51, -0.64, -1.27, 3.91, 2.4, 9.83, -0.93, 0.7, -1.12, 3.06, 0.53, -0.02, -0.78, -1.01, 3.28, 0.41, 1.96, 1.11, 1.44, -1.46, -0.27, 1.17, -1.63, -1.11, 1.79, -1.17, -6.87, -0.8, 0.04, 2.94, 1.01, -1.05, -1.23, 2.48, -1.09, 0.89, 1.36, -1.25, -0.1, -2.11, -0.34, -0.71, -0.44, -0.42, 0.77, -1.6, -0.66, 0.35, 0.75, -1.11, -0.56, -0.58, 1.02, -0.41, -0.09, -2.1]

Рисунок 3.36: Задание 3.14.3

```

sum([exp(-x[i+1]) / (x[i] + 10) for i in 1:249])

```

3.261995204084063e-5

Рисунок 3.37: Задание 3.14.4



```

for i in 1:length(y)
    if y[i] > 600
        print(i, " ", y[i], "; ")
    end
end
end

```

2) 651; 3) 850; 6) 916; 8) 814; 13) 924; 14) 907; 16) 927; 19) 769; 23) 847; 25) 673; 27) 705; 31) 868; 40) 605; 47) 639; 48) 729; 49) 744; 50) 726; 52) 638; 55) 605; 66) 738; 68) 709; 69) 912; 70) 797; 72) 681; 74) 927; 76) 937; 78) 784; 83) 731; 84) 765; 85) 757; 91) 978; 92) 799; 95) 619; 104) 718; 105) 647; 106) 913; 107) 890; 108) 907; 112) 912; 115) 898; 117) 951; 118) 811; 119) 841; 121) 999; 123) 601; 127) 978; 129) 758; 130) 789; 131) 650; 138) 837; 140) 602; 142) 973; 144) 638; 145) 820; 148) 870; 151) 634; 160) 798; 163) 835; 164) 793; 167) 942; 170) 956; 171) 689; 172) 970; 174) 745; 175) 668; 178) 620; 179) 850; 184) 986; 186) 846; 187) 678; 193) 898; 198) 878; 202) 835; 206) 626; 208) 626; 210) 800; 213) 753; 230) 667; 231) 727; 234) 917; 238) 684; 239) 633; 240) 833; 241) 628; 242) 898; 247) 968;

Рисунок 3.38: Задание 3.14.5

```

for i in 1:length(y)
    if y[i] > 600
        print(i, " ", x[i], "; ")
    end
end
end

```

2) 791; 3) 128; 6) 985; 8) 183; 13) 458; 14) 406; 16) 280; 19) 310; 23) 381; 25) 342; 27) 896; 31) 487; 40) 794; 47) 350; 48) 972; 49) 432; 50) 302; 52) 527; 55) 841; 66) 30; 68) 219; 69) 682; 70) 604; 72) 664; 74) 689; 76) 402; 78) 272; 83) 955; 84) 209; 85) 613; 91) 409; 92) 719; 95) 922; 104) 330; 105) 366; 106) 318; 107) 676; 108) 405; 112) 86; 115) 736; 117) 857; 118) 272; 119) 986; 121) 716; 123) 822; 127) 546; 129) 755; 130) 135; 131) 420; 138) 613; 140) 315; 142) 657; 144) 693; 145) 620; 148) 459; 151) 721; 160) 973; 163) 649; 164) 931; 167) 25; 170) 35; 171) 147; 172) 377; 174) 263; 175) 48; 178) 147; 179) 393; 184) 749; 186) 723; 187) 922; 193) 204; 198) 272; 202) 769; 206) 719; 208) 270; 210) 67; 213) 114; 230) 210; 231) 219; 234) 891; 238) 566; 239) 795; 240) 263; 241) 150; 242) 210; 247) 16;

Рисунок 3.39: Задание 3.14.6

```
using Statistics
vect = [round(abs(i - mean(x))^(1/2), digits=2) for i in x]
print(vect)
```

[11.4, 17.06, 19.29, 21.63, 13.08, 22.02, 7.48, 17.8, 16.31, 13.6, 18.4  
1, 18.63, 6.48, 9.69, 21.56, 14.83, 10.86, 6.93, 13.78, 22.11, 3.47, 20  
.95, 10.91, 4.35, 12.57, 19.88, 19.9, 20.54, 18.82, 12.85, 3.6, 21.26,  
17.29, 15.81, 13.89, 8.06, 6.86, 8.0, 11.27, 17.15, 20.45, 21.31, 17.17  
, 14.46, 21.12, 2.65, 12.25, 21.73, 8.24, 14.07, 19.34, 6.09, 2.82, 8.0  
, 18.47, 11.62, 22.02, 18.08, 10.86, 6.0, 14.0, 19.0, 19.54, 15.46, 21.  
42, 21.68, 13.64, 16.76, 13.49, 10.2, 13.45, 12.81, 15.75, 13.75, 19.26  
, 9.9, 16.64, 15.1, 18.14, 15.1, 8.06, 19.39, 21.33, 17.06, 10.63, 7.81  
, 14.42, 10.77, 21.24, 9.64, 9.54, 14.8, 12.89, 20.66, 20.54, 18.74, 16  
.15, 18.25, 18.79, 8.54, 12.61, 19.92, 21.21, 13.04, 11.57, 13.49, 13.2  
7, 9.74, 15.36, 15.56, 14.87, 20.35, 4.13, 16.97, 15.36, 19.34, 18.9, 1  
5.1, 22.05, 4.0, 14.7, 18.3, 17.95, 16.67, 10.82, 14.18, 6.78, 21.91, 1  
5.97, 19.1, 8.94, 6.86, 14.07, 20.05, 17.21, 21.84, 6.0, 10.63, 8.0, 13  
.6, 15.23, 12.53, 16.76, 13.89, 10.96, 11.58, 17.41, 6.4, 20.35, 9.33,  
14.87, 16.85, 11.4, 12.16, 15.39, 20.71, 12.89, 19.57, 15.33, 21.75, 3.  
61, 6.48, 12.21, 20.76, 4.13, 12.21, 21.79, 10.96, 8.06, 21.56, 18.79,  
11.09, 13.49, 15.39, 21.26, 16.34, 17.0, 18.79, 10.34, 18.11, 10.1, 22.  
27, 19.18, 15.78, 17.92, 14.93, 20.54, 20.45, 6.08, 15.23, 19.0, 20.0,  
17.2, 15.84, 11.62, 14.28, 9.27, 15.1, 14.42, 21.0, 17.94, 16.4, 11.13,  
9.8, 7.75, 14.8, 1.74, 15.16, 18.95, 20.81, 9.9, 13.08, 19.65, 11.09, 4  
.59, 19.85, 13.53, 9.75, 13.75, 19.62, 22.18, 18.36, 15.16, 20.4, 17.52  
, 7.93, 4.47, 21.38, 22.09, 17.03, 16.76, 14.28, 1.99, 19.77, 8.89, 15.  
94, 15.2, 8.13, 17.18, 15.39, 18.71, 17.03, 22.25, 11.45, 15.68, 20.93,  
22.0, 22.14, 16.25, 17.75]

Рисунок 3.40: Задание 3.14.7

```
length([i for i in y if abs(maximum(y) - i) <= 200])
```

42

Рисунок 3.41: Задание 3.14.8

```
println("Even: ", count(true for i in x if i%2==0))
println("Odd: ", count(true for i in x if i%2!=0))
```

Even: 123  
Odd: 127

Рисунок 3.42: Задание 3.14.9

```
println("Devides by 7: ", count(true for i in x if i%7==0))
```

Devides by 7: 44

Рисунок 3.43: Задание 3.14.10

```
i = sortperm(y)
sorted_x = x[i]
print(sorted_x)
```

```
[201, 463, 895, 178, 352, 435, 708, 147, 118, 698, 560, 548, 918, 914,
197, 435, 547, 126, 957, 239, 815, 296, 735, 351, 595, 436, 513, 631, 7
88, 536, 318, 407, 683, 15, 598, 421, 617, 709, 701, 222, 50, 861, 250,
916, 989, 859, 883, 427, 61, 4, 365, 100, 49, 952, 480, 41, 436, 384, 9
46, 732, 689, 373, 492, 517, 35, 254, 871, 139, 444, 103, 929, 258, 496
, 666, 807, 172, 512, 233, 766, 205, 918, 517, 968, 249, 851, 98, 686,
833, 685, 796, 270, 665, 20, 292, 587, 507, 827, 481, 268, 458, 618, 78
4, 435, 789, 927, 748, 781, 484, 115, 954, 666, 503, 437, 754, 304, 376
, 854, 620, 941, 721, 307, 464, 868, 236, 8, 153, 12, 404, 630, 602, 26
9, 329, 263, 995, 78, 564, 876, 171, 414, 521, 106, 634, 821, 382, 990,
365, 23, 319, 341, 736, 728, 296, 561, 777, 370, 126, 62, 835, 839, 623
, 739, 547, 329, 837, 822, 315, 794, 841, 922, 147, 719, 270, 150, 795,
721, 537, 693, 350, 366, 420, 791, 210, 48, 342, 922, 664, 566, 147, 89
6, 219, 330, 302, 219, 972, 955, 30, 432, 263, 114, 613, 755, 209, 310,
272, 135, 931, 604, 973, 719, 67, 272, 183, 620, 263, 649, 769, 613, 98
6, 723, 381, 128, 393, 487, 459, 272, 676, 736, 204, 210, 406, 405, 682
, 86, 318, 985, 891, 458, 280, 689, 402, 25, 857, 35, 16, 377, 657, 409
, 546, 749, 716]
```

Рисунок 3.44: Задание 3.14.11

```
top_10 = partialsort(x, 1:10, rev=true)
print(top_10)
```

```
[995, 990, 989, 986, 985, 973, 972, 968, 957, 955]
```

Рисунок 3.45: Задание 3.14.12

```
temp_x = Set(x)
temp_x = collect(temp_x)
print(temp_x)
```

```
[402, 35, 689, 719, 30, 723, 941, 67, 219, 929, 319, 272, 871, 657, 153
, 115, 547, 420, 404, 861, 139, 665, 365, 4, 405, 86, 263, 183, 376, 63
0, 12, 676, 23, 302, 41, 171, 487, 895, 857, 666, 406, 517, 126, 955, 1
72, 685, 382, 103, 598, 280, 521, 833, 837, 307, 755, 366, 414, 100, 35
2, 484, 20, 721, 754, 329, 135, 512, 222, 732, 409, 481, 918, 972, 891,
249, 381, 839, 916, 209, 957, 236, 250, 859, 649, 78, 560, 990, 436, 25
8, 564, 735, 8, 435, 789, 201, 427, 604, 507, 686, 464, 270, 777, 620,
968, 548, 602, 868, 716, 480, 432, 709, 634, 254, 496, 566, 683, 595, 3
15, 269, 114, 794, 827, 268, 985, 197, 748, 444, 784, 233, 342, 954, 21
0, 239, 736, 698, 393, 946, 351, 285, 739, 821, 62, 682, 437, 150, 766,
896, 587, 952, 851, 989, 791, 854, 623, 98, 204, 118, 377, 883, 537, 91
4, 25, 310, 384, 292, 796, 973, 50, 318, 728, 296, 304, 788, 769, 147,
927, 458, 16, 370, 341, 807, 49, 664, 693, 708, 749, 503, 106, 61, 841,
876, 513, 995, 631, 128, 350, 835, 815, 463, 781, 561, 922, 421, 373, 4
92, 407, 546, 613, 178, 617, 536, 459, 822, 931, 986, 795, 701, 618, 48
, 15, 330]
```

Рисунок 3.46: Задание 3.14.13

Задание 4. Квадраты целых чисел от 1 до 100.

```
squares = [i^2 for i in 1:100]
print(squares)
```

[1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400, 441, 484, 529, 576, 625, 676, 729, 784, 841, 900, 961, 1024, 1089, 1156, 1225, 1296, 1369, 1444, 1521, 1600, 1681, 1764, 1849, 1936, 2025, 2116, 2209, 2304, 2401, 2500, 2601, 2704, 2809, 2916, 3025, 3136, 3249, 3364, 3481, 3600, 3721, 3844, 3969, 4096, 4225, 4356, 4489, 4624, 4761, 4900, 5041, 5184, 5329, 5476, 5625, 5776, 5929, 6084, 6241, 6400, 6561, 6724, 6889, 7056, 7225, 7396, 7569, 7744, 7921, 8100, 8281, 8464, 8649, 8836, 9025, 9216, 9409, 9604, 9801, 10000]

Рисунок 3.47: Квадраты целых чисел от 1 до 100

#### Задание 5. Операции с простыми числами.

```
using Primes
myprimes = primes(prime(168))
print(myprimes)
```

[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499, 503, 509, 521, 523, 541, 547, 557, 563, 569, 571, 577, 587, 593, 599, 601, 607, 613, 617, 619, 631, 641, 643, 647, 653, 659, 661, 673, 677, 683, 691, 701, 709, 719, 727, 733, 739, 743, 751, 757, 761, 769, 773, 787, 797, 809, 811, 821, 823, 827, 829, 839, 853, 857, 859, 863, 877, 881, 883, 887, 907, 911, 919, 929, 937, 941, 947, 953, 967, 971, 977, 983, 991, 997]

```
prime(89)
```

461

```
print(myprimes[89:99])
```

[461, 463, 467, 479, 487, 491, 499, 503, 509, 521, 523]

Рисунок 3.48: Операции с простыми числами

#### Задание 6. Вычисления.

```
summa = 0
for i in 10:100
    summa += i^3 + 4*i^2
end
print(summa)
```

26852735

```
M = 25
summa = 0
for i in 1:M
    summa += (2^i)/i + (3^i)/(i^2)
end
print(summa)
```

2.1291704368143802e9

```
N = 38
summa = 1
a_n = 1
for i in 2:2:N
    a_n *= i/(i+1)
    summa += a_n
end
print(summa)
```

6.976346137897618

Рисунок 3.49: Вычисления

## 4 Выводы

Изучены несколько структур данных, реализованных в Julia, их применение и операции над ними для решения задач.

# Список литературы

- [1] Julia Language Documentation. *Julia 1.5 Documentation*. 2020. URL: <https://docs.julialang.org/en/v1/>.